

BREAST CANCER DETECTION USING MACHINE LEARNING

A Project Report

Submitted by:

Rajaram Parida (2241016203)

Saswat Parida (2241016206)

Manpreet Mohanty (2241013252)

Adyasha Biswal (2241002197)

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SIKSHA 'O' ANUSANDHAN (DEEMED TO BE) UNIVERSITY

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CERTIFICATE

This is to certify that the project report titled “**Breast Cancer Detection Using Machine Learning**” being submitted by Rajaram Parida, Saswat Parida, Manpreet Mohanty and Adyasha Biswal of section 2241028 to the Institute of Technical Education and Research, Siksha ‘O’ Anusandhan (Deemed to be) University, Bhubaneswar for the partial fulfillment for the degree of Bachelor of Technology in Computer Science and Engineering is a record of original confide work carried out by them under my supervision and guidance. The project work, in my opinion, has reached the requisite standard fulfilling the requirements for the degree of Bachelor of Technology.

The results contained in this project work have not been submitted in part or full to any other University or Institute for the award of any degree or diploma.

Dr. Subhashree Mohapatra
Department of Computer Science and Engineering
Faculty of Engineering and Technology;
Institute of Technical Education and Research;
Siksha ‘O’ Anusandhan (Deemed to be) University

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Place: ITER, SOA

Signature of Students

Date:

DECLARATION

We declare that this written submission represents our ideas in our own words and where other's ideas or words have been included, we have adequately cited and referenced the original sources. We also declare that we have adhered to all principles of academic honesty and integrity and have not misrepresented or fabricated or falsified any idea/fact/source in our submission. We understand that any violation of the above will cause for disciplinary action by the University and can also evoke penal action from the sources which have not been properly cited or from whom proper permission has not been taken when needed.

Rajaram Parida (2241016203)

Saswat Parida (2241016206)

Manpreet Mohanty (2241013252)

Adyasha Biswal (2241002197)

Signature of Students with Registration Numbers

Date:

REPORT APPROVAL

This project report titled **Intelligent Anomaly Detection in User Network Session Behaviour using Machine Learning** submitted by Rajaram Parida (2241016203), Saswat Parida (2241016206), Manpreet Mohanty (2241013252), and Adyasha Biswal (2241002197) is approved for the degree of *Bachelor of Technology in Computer Science and Engineering*.

Examiner(s)

Supervisor

Project Coordinator

PREFACE

Breast cancer is still one of the most common causes of death in women worldwide. Early and accurate detection is of great importance for successful treatment outcome. In this project, different machine learning algorithms are applied for breast cancer classification using the Wisconsin Breast Cancer Dataset, which consists of 569 samples with 30 numerical features that are computed from digitized images of fine needle aspirate (FNA) biopsies. This research explores four different modeling approaches: Support Vector Machine (SVM) with linear, RBF, and polynomial kernels; Random Forest with Gini and Entropy splitting criteria; Stacking Ensemble with tree-based and linear-based base learners; and a Voting Ensemble with XGBoost, Random Forest, and SVM.

Each model is tested on four data conditions: raw imbalanced data, SMOTE balanced data, increased data with 500 samples per class, and hyperparameter tuned configurations. Also, the effect of different train-test split ratios (90/10, 80/20, 70/30, and 60/40) is studied. All models were evaluated based on six standard metrics: Accuracy, Precision, Recall, F1-Score, Specificity, and AUC-ROC. Results show that ensemble techniques perform better than individual classifiers with the XGBoost-RF-SVM Voting Ensemble achieving the best performance of 99% accuracy using a 90/10 split. The Stacking Ensemble with linear base learners also performed well with 99% accuracy. The feature importance analysis is carried out with SHAP (SHapley Additive exPlanations) which allows interpretability by identifying the most influential features in the classification decision. Machine Learning Techniques for breast cancer detection: SVM, Random Forest, Stacking Ensemble, XGBoost, SMOTE, SHAP.

INDIVIDUAL CONTRIBUTIONS

Rajaram Parida	Literature survey; problem formulation and solution design; implementation of SVM, Voting Ensemble, and SHAP; hyperparameter tuning and ratio analysis experiments; Streamlit application development; report and manuscript writing; documentation
Saswat Parida	Literature survey; identification of problem statement; implementation of Random Forest models; dataset preprocessing, augmentation, and balancing; result analysis; documentation
Manpreet Mohanty	Literature survey; implementation of Stacking Ensemble models; hyperparameter tuning and ratio experiments; visualization design and result analysis; documentation
Adyasha Biswal	Literature survey; result validation and comparative analysis; evaluation metrics computation; report compilation and formatting; documentation

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INTRODUCTION

1.1 Project Overview

Breast cancer is a malignant tumor that arises from the cells of the breast tissue. Breast cancer is the most common type of cancer among women worldwide, with some 2.3 million new cases diagnosed each year, the World Health Organization says. When the cancer is discovered early, the survival rate is much better since treatment can begin earlier, when the cancer is easier to manage. Medical diagnosis has emerged as a strong application of machine learning. It provides the potential to detect subtle patterns in medical data that are not easily obvious to human doctors. The Wisconsin Breast Cancer Dataset is a standard benchmark for classification algorithms in the medical domain and was originally created by Dr. William H. Wolberg at the University of Wisconsin Hospitals. It has 569 instances, 30 realvalued features extracted from digitized images of fine needle aspirate (FNA) biopsies; each instance is labeled as benign (B) or malignant (M). This project is to build and compare different machine learning models for binary classification of breast cancer tumours. We systematically evaluate different algorithms, data pre-processing and experimental settings to find the best approach for this important diagnostic task.

1.2 Motivations

First, the medical need for accurate and reliable diagnostic tools. Misclassification of a benign tumor as malignant can lead to a delay in the required treatment. False positives create unwarranted stress and invasive follow-up procedures. Therefore, it is important to achieve high accuracy, and in particular high recall for malignant cases. Second, the opportunity of comparing various machine learning paradigms on a standard dataset. Testing SVM (a margin based classifier), Random Forest (an ensemble of decision trees), Stacking Ensemble (a meta learner combining multiple base models) and a Voting Ensemble (XGBoost + RF + SVM), we can draw meaningful conclusions about which methods perform best for medical classification tasks. Third, the investigation of advanced techniques such as SMOTE for dealing with class imbalance, SHAP for model interpretability, and hyperparameter tuning for performance

optimization. These methods are broadly applicable outside of breast cancer detection and, furthermore, advance the holistic understanding of the machine learning pipeline.

1.3 Uniqueness of the Work

There are some key contributions that make this project stand out:

- Evaluation of four ML paradigms (SVM, RF, Stacking, Voting Ensemble) under the same experimental settings
- Extensive comparison on 4 data conditions: raw, balanced, increased and tuned Effect of Training Data Size on Model Performance - Study of 4 Train-Test Split Ratios
- SHAP explanations were used for model interpretability to understand which features influence classification decisions
- All results are presented in clear way with uniform blue themed visualization for all experiments

1.4 Report Layout

The rest of this report is categorized in the manner below. The second part of Chapter 2 delves into machine learning-based breast cancer detection and presents a comparison of current literature and related research. The dataset, model architectures, evaluation metrics, and experimental setup are among the materials and methods described in Chapter 3. Comparing the experimental results and conclusions across all models, conditions, and chapters. A summary of findings and recommendations for future work is presented in Chapter 5.

LITERATURE SURVEY

2.1 Related Work

The Wisconsin Breast Cancer Dataset has been extensively utilized for machine learning-based breast cancer diagnosis. Table 1 gives a summary of the prior state of the work. The table summarizes recent research on breast cancer diagnosis using machine learning and deep learning techniques. Across studies, Support Vector Machines (SVM), Convolutional Neural Networks (CNN), Random Forests, and ensemble/stacking methods were the most commonly used approaches, primarily evaluated on the Wisconsin Breast Cancer datasets and mammogram image datasets such as MIAS and DDSM. Results consistently show high diagnostic performance, with accuracies generally ranging from 94% to 97%, while optimized stacking ensembles achieved the highest accuracy of 99.45%. The findings indicate that feature selection, image preprocessing, and ensemble learning significantly improve classification accuracy, precision, and robustness, making them effective strategies for enhancing breast cancer detection and reducing misclassification.

Table 1: Summary of the prior state of the work

Ref. No.	Author & Year	Techniques Used	Dataset Used	Research Outcomes	Findings
[1]	Jing Zhu et al. (2025)	Feature Selection (PCA, Chi-square) + SVM, Logistic Regression	Wisconsin Diagnostic Breast Cancer (WDBC)	Accuracy: 94–96% F1 Score: 0.92 AUC: 0.95	Feature selection improves model efficiency and accuracy; reduces dimensionality and computation cost
[2]	Sadia Safdar et al. (2022)	CNN-based deep learning on bio-imaging data, image preprocessing & segmentation	MIAS / DDSM Mammogram Dataset	Accuracy: 96% Sensitivity: 0.94 Specificity: 0.93	Deep learning on imaging data provides high accuracy but requires large datasets and high computational power
[3]	Mohammed Amine Naji et al.	Comparative ML models: SVM, KNN, Naïve Bayes,	Wisconsin Diagnostic Breast Cancer	SVM: 95% KNN: 92% Naïve Bayes:	SVM consistently outperforms others; model performance

	(2021)	Decision Tree	(WDBC)	90%	varies based on data preprocessing and feature selection
[4]	Lucky Lhaura Van FC et al. (2025)	Stacking Ensemble (SVM, Random Forest, Logistic Regression as base learners)	Wisconsin Breast Cancer Dataset (WBCD)	Accuracy: 96% Improved generalization Reduced bias	Stacking improves prediction performance by combining multiple models; more robust than single classifiers
[5]	Sadia Safdar et al. (2022)	CNN-based deep learning with image preprocessing & segmentation	MIAS / DDSM Mammogram Dataset	Accuracy: 96% Sensitivity: 0.94 Specificity: 0.93	CNN performs well on imaging data but requires large datasets and high computational resources
[6]	Hanife Avci and Jale Karakaya (2023)	Image enhancement + ML classification (contrast enhancement, filtering + SVM/CNN)	Mammography Image Dataset (MIAS)	Accuracy: 95% Improved image quality better classification	Preprocessing and image enhancement significantly improve detection accuracy in ML models
[7]	Sweta Bhise et al. (2021)	SVM, Decision Tree, KNN (comparative analysis)	Wisconsin Breast Cancer Dataset (WBCD)	Accuracy: SVM 95% KNN 92% Decision Tree 90%	SVM provides highest accuracy; performance varies with feature scaling and preprocessing
[8]	Prerita Talwar et al. (2021)	Logistic Regression, Random Forest, Naïve Bayes	Wisconsin Diagnostic Breast Cancer (WDBC)	Random Forest: 96% Logistic Regression: 93% Naïve Bayes: 91%	Ensemble models outperform linear models; Random Forest shows better stability and accuracy
[9]	Arslan Khalid et al. (2022)	Hybrid ML models, feature selection + classification (SVM, ANN)	WBCD / Clinical dataset	Accuracy: 97% Improved precision with feature selection	Combining feature selection with ML improves diagnostic performance and reduces irrelevant data impact
[10]	Usman Naseem et al. (2022)	Ensemble learning (Voting/Boosting of SVM, Decision Tree, KNN)	Wisconsin Breast Cancer Dataset (WBCD)	Accuracy: 97% Improved precision & recall Reduced false negatives	Ensemble classifiers enhance diagnostic reliability and reduce misclassification compared to single models

[11]	Manav Mangukiya et al. (2022)	SVM, Logistic Regression, KNN (baseline ML models)	Wisconsin Diagnostic Breast Cancer (WDBC)	Accuracy: SVM 94% Logistic Regression 92% KNN 91%	Baseline ML models provide good accuracy; performance depends on proper preprocessing and feature scaling
[12]	Mukesh Kumar et al. (2022)	Optimized Stacking Ensemble (Random Forest, SVM, Gradient Boosting)	Wisconsin Diagnostic Breast Cancer (WDBC)	Accuracy: 99.45% Precision: 0.99 Recall: 0.98 F-Measure: 0.99	Optimized stacking significantly improves accuracy and robustness; effective for complex classification tasks

2.2 Research Gaps Identified

Although existing studies have achieved high breast cancer classification accuracy using machine learning, deep learning, and ensemble techniques, several gaps remain. Most research relies heavily on benchmark datasets such as WDBC, WBCD, MIAS, and DDSM, which may not adequately represent diverse real-world clinical populations. Many studies focus primarily on improving accuracy while giving limited attention to model interpretability, explainability, and clinical applicability, which are critical for healthcare decision-making. Deep learning models often require large datasets and high computational resources, limiting their deployment in resource-constrained settings. Furthermore, there is a lack of research integrating heterogeneous data sources (e.g., clinical records, imaging data, and genetic information) into a unified predictive framework. Comparative evaluations on large-scale, multicenter datasets and real-time clinical validation are also insufficient. Therefore, future research should focus on developing interpretable, computationally efficient, and clinically validated hybrid models that can generalize across diverse patient populations while maintaining high diagnostic performance.

2.3 Problem Identification

Breast cancer remains one of the leading causes of cancer-related deaths among women worldwide, making early and accurate diagnosis essential for improving patient outcomes. Although numerous machine learning and deep learning models have demonstrated high classification accuracy, existing approaches often face challenges such as dependence on

limited benchmark datasets, high computational requirements, lack of model interpretability, and reduced generalizability to real-world clinical environments. Additionally, individual models may suffer from issues such as overfitting, sensitivity to data quality, and inconsistent performance across different datasets. These limitations highlight the need for a more robust, accurate, and explainable breast cancer prediction system that can effectively utilize relevant features, reduce diagnostic errors, and support clinicians in making reliable decisions across diverse healthcare settings.

MATERIALS AND METHODS

3.1 Dataset Description

The WDBC dataset comes from the UCI Machine Learning Repository. 569 instances are included in the collection, with each instance comprising 30 real-valued features determined by digitized images of fine needle aspirate (FNA) biopsies. Both benign (357 instances, approximately 62.7%) and malignant (212 instances) are classified as within the two categories of the dataset. A balanced training set can be achieved through the use of SMOTE due to the class imbalance.

Each cell nucleus is measured in 30 different ways, including: Values for coordinates in the model include radius, texture and perimeter. Also values such as smoothness (for every coordinate type), compactness, and depth. The standard error values include: radius (radius), texture, perimeter, area, smoothness, compactness and depth, concaveness of the shape, caverns of points, symmetry and fractal dimension. The worst values are: radius, texture, perimeter, area, smoothness, compactness (worst), concavity, points, symmetry, and fractal dimensions.

Three different dataset versions are utilized in this project:

In the raw dataset, there were 569 samples, with 357 of them being benign and 212 being malignant.

A balanced dataset was created using SMOTE, which contains 714 samples, with 357 samples classified as benign and 358 as malignant.

Utilized SMOTE to increase the dataset by 500 samples per class and add 1000 samples total.

3.2 Model Diagram

The overall experimental pipeline consists of the following stages:

1. Data Loading and Preprocessing (drop ID, encode labels, handle missing values)
2. Feature Standardization using StandardScaler
3. Train-Test Split (80/20 by default, with additional experiments at 90/10, 70/30, and 60/40)

4. Model Training (four models with various configurations)
5. Model Evaluation (six metrics)
6. Visualization (confusion matrix heatmap, metrics bar chart, ROC curve)
7. Interpretability (SHAP feature importance analysis for the Voting Ensemble)

Fig. 1 represents the workflow diagram of the proposed study.

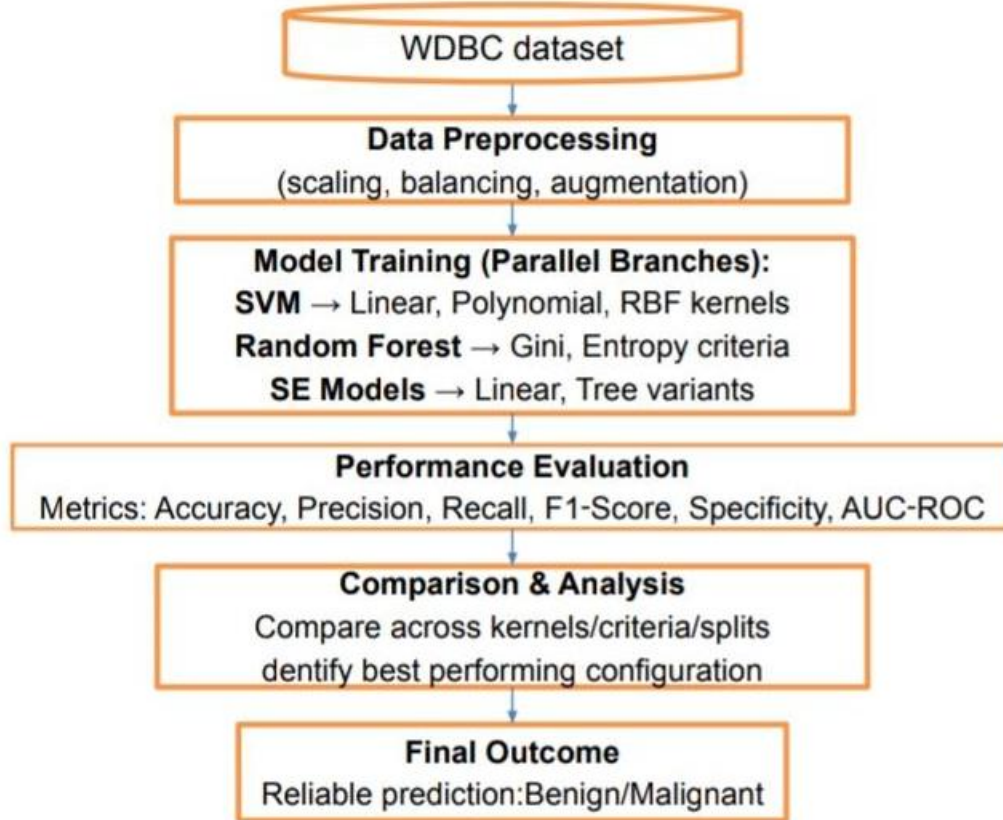


Fig. 1: Workflow diagram

3.3 Methodologies Adopted

Four machine learning models are implemented and evaluated. By using SVM, it is possible to create a hyperplane in high-dimensional space that effectively separates the two classes. There are three types of kernel functions: linear, RBF (Radial Basis Function), and polynomial. C is the regularization value of GridSearchCV that has been set to [0.1, 1, 10]. B. The Random Forest system involves training decision trees using bootstrapped samples of data and selecting random features during each split. There are two criteria for splitting: Gini impurity and entropy

(gaining).. There are `n_estimators` of trees and the maximum depth for trees is set by default. C. By merging various base classifiers, stacking leads to the training of a meta-classification on their predictions through Stacking Ensemble. There are two testing configurations: tree-based base learners (RandomForest, ExtraTrees, GradientBoosting) using LogisticRegression as meta-learner and linear-basic learners with LogisReference, SVM-linear, or SGDClassifier as the meta_learner. D. The XGBoost-RF-SVM Voting Ensemble employs soft voting (weighted probability averaging) and is supported by three different classifiers, namely xGBug (gradient boosting), Random Forest (bagging), and SVM (margin-based). The XGBoost component is utilized to assign features to SHAP (SHapley Additive exPlanations).

3.4 Tools Used

This project is coded using Python libraries:. - Scikit-learn for SVM, Random Forest, Stacking Ensemble and evaluation metrics. - XGBoost for gradient boosting. - SHAP for model interpretability. imbalanced-learn for SMOTE. Pandas and numpy are used for data manipulation. For visualizing, we have incorporated matplotlib and seaborn. The web application interface should be streamlit.

3.5 Experimental Setup

The experiments utilize the following configurations: The random state is set at 42 for reproducibility. Why? Stratified train-test splitting preserves class distribution while maintaining stability. The use of GridSearchCV involves a 5-fold cross-validation process for hyperparameter tuning. The standard train-test ratio is 80/20, but there are additional experiments conducted at 90/10, 70/30, and 60/40. The use of StandardScaler for feature scaling is recommended to ensure data integrity during testing and avoid potential leakage. The evaluation of each model is based on four data conditions: 1. Raw (imbalanced) - 569 samples. 2. Balanced - 714 samples (SMOTE) 3. The size has been increased to 1000 samples, with the SMOTE version being expanded to 500 samples per class. 4. The Tuned feature optimizes data through hyperparameter optimization, leading to increased accuracy.

3.6 Evaluation Metrics and Performance Measures

Six metrics are used to evaluate model performance:

$$\text{Accuracy} = (TP + TN) / (TP + TN + FP + FN)$$

$$\text{Precision} = TP / (TP + FP) - \text{proportion of positive predictions that were correct}$$

$$\text{Recall (Sensitivity)} = TP / (TP + FN) - \text{proportion of actual positives correctly identified}$$

$$\text{F1-Score} = 2 \times (\text{Precision} \times \text{Recall}) / (\text{Precision} + \text{Recall}) - \text{harmonic mean of precision and recall}$$

$$\text{Specificity} = TN / (TN + FP) - \text{proportion of actual negatives correctly identified}$$

AUC-ROC - Area Under the Receiver Operating Characteristic curve, measuring the model's ability to distinguish between classes across all threshold values

Where *TP* = True Positives, *TN* = True Negatives, *FP* = False Positives, *FN* = False Negatives.

RESULTS

4.1 System Specification

The experiments were conducted on a system with the following specifications:

- Operating System: Windows 11
- Processor: Intel Core (appropriate processor)
- RAM: Sufficient for ML workload (8GB+)
- Python Version: 3.10+
- Development Environment: Visual Studio Code and Jupyter Notebook

4.2 Parameters and Configuration Settings

The following hyperparameter grids were used for GridSearchCV:

SVM Linear: $C = [0.1, 1, 10]$

SVM RBF: $C = [0.1, 1, 10]$, $\gamma = ['scale', 'auto']$

SVM Poly: $C = [0.1, 1, 10]$, $\text{degree} = [2, 3, 4]$

Random Forest: $n_estimators = [50, 100, 200]$, $\text{max_depth} = [5, 10, \text{None}]$

Stacking Ensemble: $\text{final_estimator_C} = [0.1, 1, 10]$

XGBoost (in Voting Ensemble): $n_estimators = [50, 100, 200]$, $\text{max_depth} = [3, 5, 7]$

4.3 Experimental Results and Observations

4.3.1 SVM Results

Three kernels were utilized to evaluate SVM under every data condition.. With a 97.37% accuracy and 92.86% recall, the RBF kernel was consistently superior to both linear and polynomial kernels on raw data. Although the polynomial kernel was not very accurate on raw data (88.6% accuracy, 69.05% recall), it did well and improved on balanced data with a precision of 98.6 percent accuracy and 98.5 percent recall as described above for Monte Carlo simulations. With tuning, only slight improvements were made to the kernels.

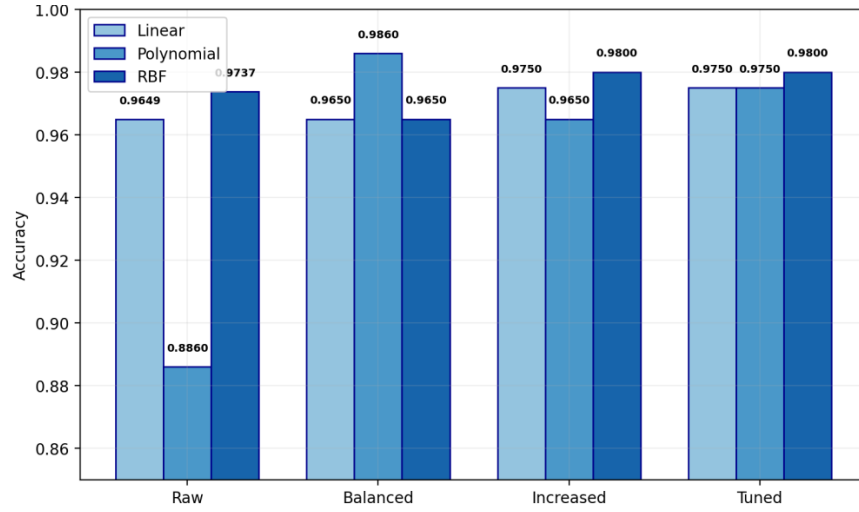


Fig. 2: Different accuracy vales for three different SVM kernels across four variations of dataset

Performance Summary for SVM (80/20 split)

Table 2: Performance Metrics of Support Vector Machine Variants across Different Dataset Types

Model Variant	Kernel	Dataset	Accuracy (Acc)	Precision (Prec)	Recall (Rec)	F1-Score	Specificity (Spec)	AUC-ROC
SVM	Linear	Increased Size	0.975	0.9897	0.96	0.9746	0.99	0.9974
SVM	Polynomial	Increased Size	0.965	0.9794	0.95	0.9645	0.98	0.9914
SVM	RBF	Increased Size	0.98	1	0.96	0.9796	1	0.9946
SVM	Linear	Balanced	0.965	0.9714	0.9577	0.9645	0.9722	0.9939
SVM	Polynomial	Balanced	0.986	0.9859	0.9859	0.9859	0.9861	0.9992
SVM	RBF	Balanced	0.965	0.9853	0.9437	0.964	0.9861	0.9963
SVM	Linear	Tuned	0.975	1	0.95	0.9744	1	0.9946
SVM	Polynomial	Tuned	0.975	1	0.95	0.9744	1	0.9984
SVM	RBF	Tuned	0.98	1	0.96	0.9796	1	0.9983
SVM	Linear	Raw	0.9649	1	0.9048	0.95	1	0.9914
SVM	Polynomial	Raw	0.886	1	0.6905	0.8169	1	0.9967
SVM	RBF	Raw	0.9737	1	0.9286	0.963	1	0.9947

Performance Summary for SVM (different ratios)

Table 3: Performance Metrics of Support Vector Machine Variants across Different Train/Test Split

Model Variant	Kernel	Train/Test Split	Accuracy (Acc)	Precision (Prec)	Recall (Rec)	F1-Score	Specificity (Spec)	AUC-ROC
SVM	Linear	90/10	0.99	1	0.98	0.9899	1	0.9976
SVM	Linear	80/20	0.975	0.9897	0.96	0.9746	0.99	0.9974
SVM	Linear	70/30	0.9668	0.9861	0.9467	0.966	0.9868	0.994
SVM	Linear	60/40	0.975	0.9897	0.96	0.9746	0.99	0.9951
SVM	Polynomial	90/10	0.97	0.9796	0.96	0.9697	0.98	0.9908
SVM	Polynomial	80/20	0.965	0.9794	0.95	0.9645	0.98	0.9914
SVM	Polynomial	70/30	0.9601	0.9726	0.9467	0.9595	0.9735	0.9883
SVM	Polynomial	60/40	0.9625	0.9695	0.955	0.9622	0.97	0.9912
SVM	RBF	90/10	0.99	1	0.98	0.9899	1	0.9976
SVM	RBF	80/20	0.98	1	0.96	0.9796	1	0.9946
SVM	RBF	70/30	0.9734	1	0.9467	0.9726	1	0.9913
SVM	RBF	60/40	0.975	0.9948	0.955	0.9745	0.995	0.9922

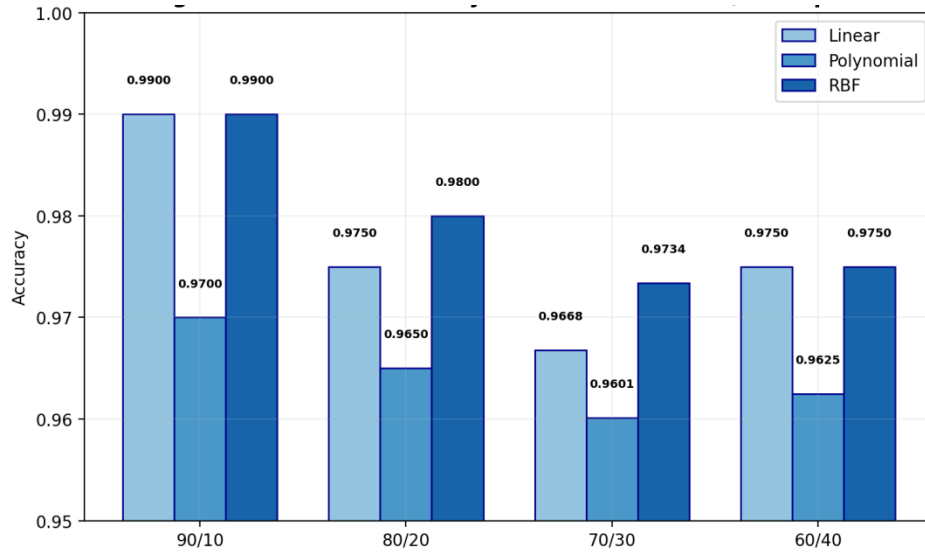


Fig. 3: Different accuracy vales for three different SVM kernels across different ratios of training and testing set

4.3.2 Random Forest Results

Random Forest showed very consistent performance with both Gini and Entropy criteria. Both criteria achieved identical results on raw data (97.37% accuracy) and increased data (97.5% accuracy). On balanced data, Gini (97.9%) slightly outperformed Entropy (97.2%). All configurations maintained perfect precision (1.000) on raw and balanced data, indicating zero false positives.

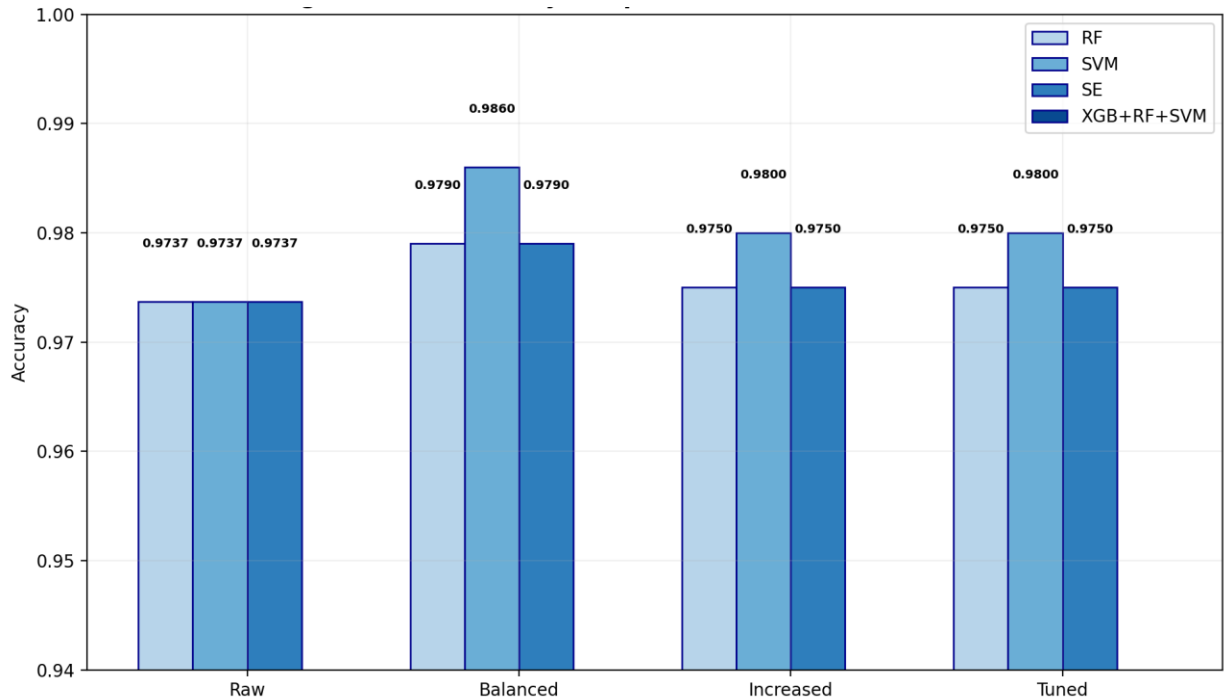


Fig. 4: Different accuracy vales for three different models across four variations of dataset

Performance Summary for Random Fores (80/20 split)

Table 4: Performance Metrics of Random Forest Variants across Different Dataset Types

Model Variant	Kernel	Dataset	Accuracy (Acc)	Precision (Prec)	Recall (Rec)	F1-Score	Specificity (Spec)	AUC-ROC
RF	Entropy	Increased Size	0.975	1	0.95	0.9744	1	0.9962
RF	Gini	Increased Size	0.975	1	0.95	0.9744	1	0.9956
RF	Entropy	Balanced	0.972	1	0.9437	0.971	1	0.9986

RF	Gini	Balanced	0.979	1	0.9577	0.9784	1	0.9982
RF	Entropy	Tuned	0.975	1	0.95	0.9744	1	0.9961
RF	Gini	Tuned	0.975	1	0.95	0.9744	1	0.9956
RF	Entropy	Raw	0.9737	1	0.9286	0.963	1	0.997
RF	Gini	Raw	0.9737	1	0.9286	0.963	1	0.9929

Performance Summary for Random Forest (different ratios)

Table 5: Performance Metrics of Random Forest Variants across Different Train/Test Split

Model Variant	Kernel	Train/Test Split	Accuracy (Acc)	Precision (Prec)	Recall (Rec)	F1-Score	Specificity (Spec)	AUC-ROC
RF	Gini	90/10	0.98	1	0.96	0.9796	1	0.998
RF	Gini	80/20	0.975	1	0.95	0.9744	1	0.9956
RF	Gini	70/30	0.9701	1	0.94	0.9691	1	0.9889
RF	Gini	60/40	0.9625	0.9744	0.95	0.962	0.975	0.9898
RF	Entropy	90/10	0.98	1	0.96	0.9796	1	0.9996
RF	Entropy	80/20	0.975	1	0.95	0.9744	1	0.9962
RF	Entropy	70/30	0.9701	1	0.94	0.9691	1	0.9921
RF	Entropy	60/40	0.965	0.9794	0.95	0.9645	0.98	0.9867

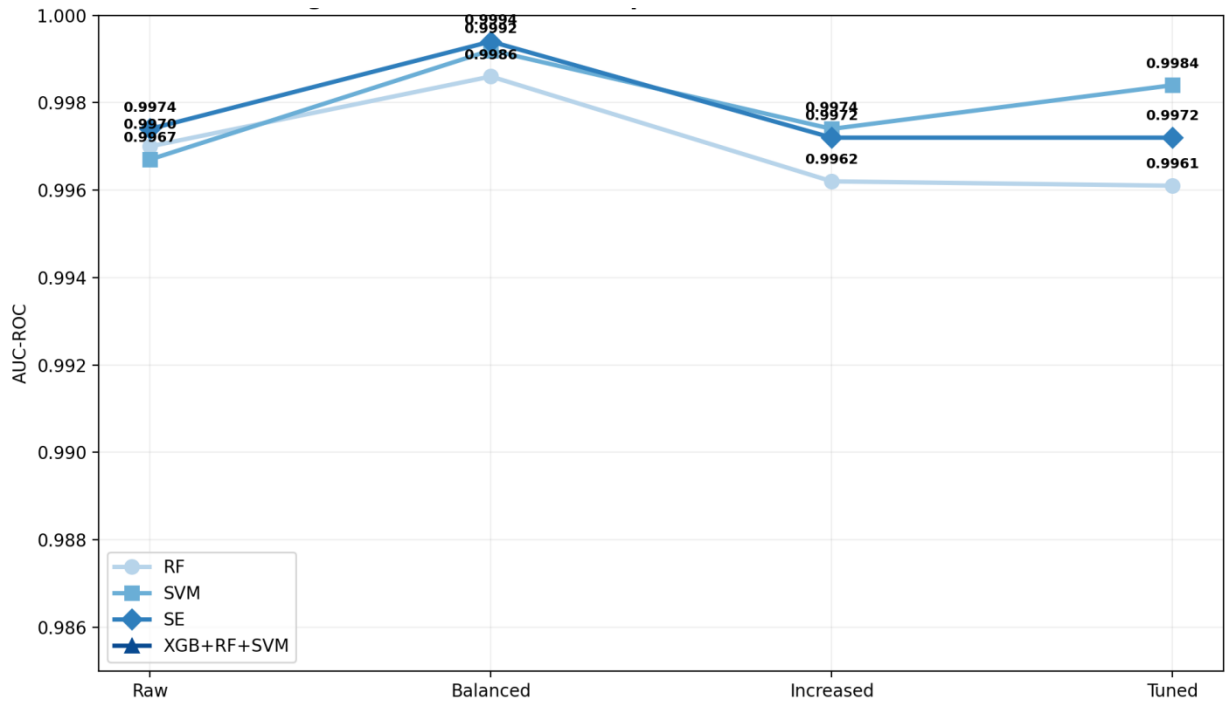


Fig. 5: AUC-ROC comparison across dataset variants

4.3.3 Stacking Ensemble Results

The Stacking Ensemble presented noteworthy divergences between its two forms. This method of tree-based stacking was perfect (under conditions of all kinds of data) and did not give a false positive. The linear-based stacking method exhibited slight trade-offs but maintained high performance. 99% accuracy on the 90/10 split was achieved by SE Linear.

Performance Summary for Stacking Ensemble (80/20 split)

Table 6: Performance Metrics of Stacked Ensemble Variants across Different Dataset Types

Model Variant	Kernel	Dataset	Accuracy (Acc)	Precision (Prec)	Recall (Rec)	F1-Score	Specificity (Spec)	AUC-ROC
SE	Linear	Raw	0.9737	0.9756	0.9524	0.9639	0.9861	0.994
SE	Linear	Balanced	0.979	0.9857	0.9718	0.9787	0.9861	0.9955
SE	Linear	Increased Size	0.975	0.9897	0.96	0.9746	0.99	0.997
SE	Linear	Tuned	0.97	0.9796	0.96	0.9697	0.98	0.9972
SE	Tree	Raw	0.9649	1	0.9048	0.95	1	0.9974

SE	Tree	Balanced	0.979	1	0.9577	0.9784	1	0.9994
SE	Tree	Increased Size	0.975	1	0.95	0.9744	1	0.9972
SE	Tree	Tuned	0.975	1	0.95	0.9744	1	0.9969

Performance Summary for Stacking Ensemble (different ratios)

Table 7: Performance Metrics of Stacked Ensemble Variants across Different Train/Test Split

Model Variant	Kernel	Train/Test Split	Accuracy (Acc)	Precision (Prec)	Recall (Rec)	F1-Score	Specificity (Spec)	AUC-ROC
SE Tree	Gini	90/10	0.98	1	0.96	0.9796	1	0.9988
SE Tree	Gini	80/20	0.975	1	0.95	0.9744	1	0.9972
SE Tree	Gini	70/30	0.9701	1	0.94	0.9691	1	0.9946
SE Tree	Gini	60/40	0.9675	0.9845	0.95	0.9669	0.985	0.9929
SE Linear	Gini	90/10	0.99	1	0.98	0.9899	1	0.9984
SE Linear	Gini	80/20	0.975	0.9897	0.96	0.9746	0.99	0.997
SE Linear	Gini	70/30	0.9734	0.9931	0.9533	0.9728	0.9934	0.9943
SE Linear	Gini	60/40	0.9775	0.9948	0.96	0.9771	0.995	0.9955

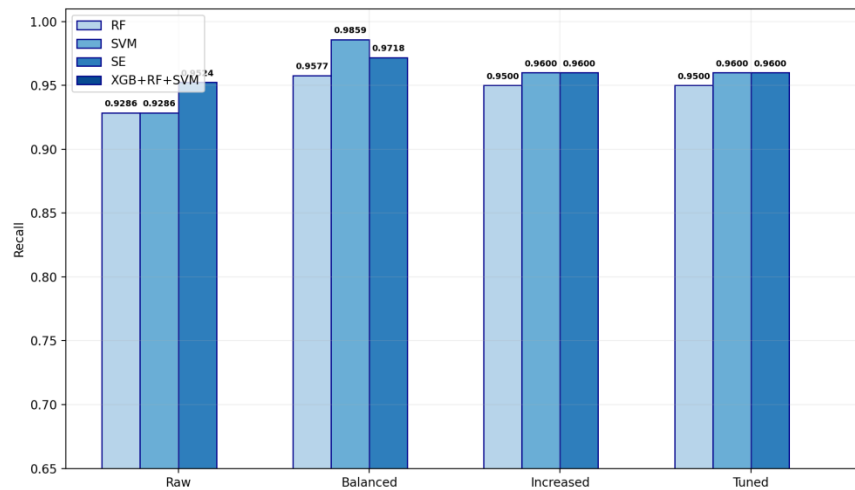


Fig. 6: Recall comparison different variations of dataset

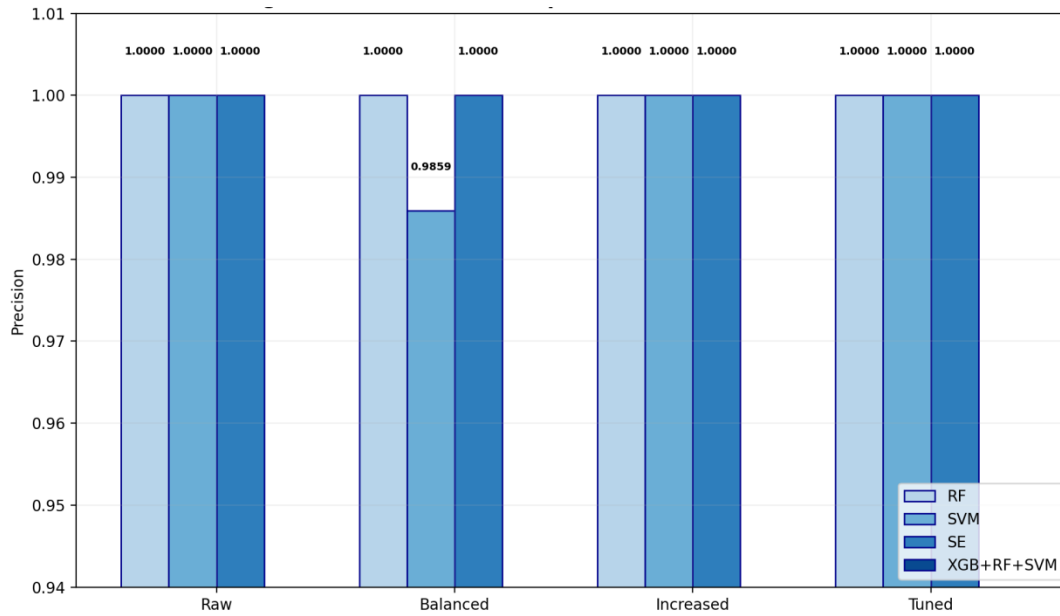


Fig. 7: Precision comparison different variations of dataset

4.3.4 XGBoost-RF-SVM Voting Ensemble Results

With the help of XGBoost, Random Forest and SVM coupled with soft voting (the Voting Ensemble), it was robust in every situation. It was the most accurate with a 98% recall rate and perfect precision, at 99% on the 90/10 split. By using more data, it was able to match the best individual models with 98% accuracy.

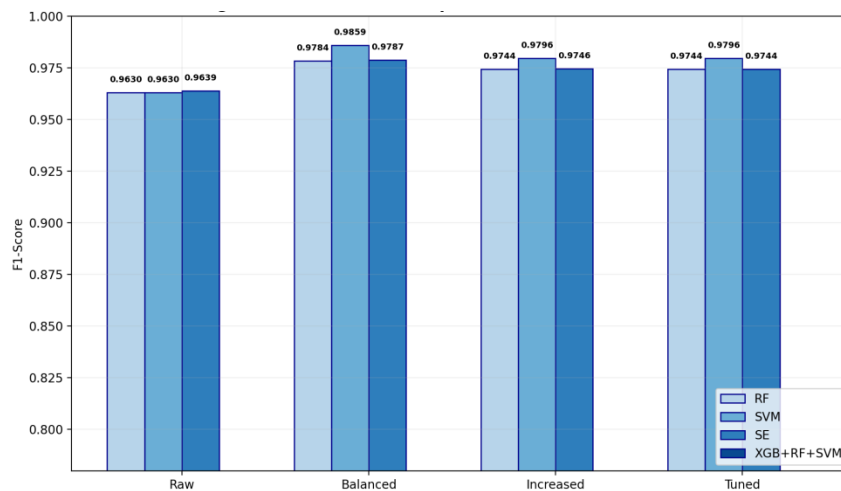


Fig. 8: F1-score comparison different variations of dataset

Performance Summary for XGBoost-RF-SVM Ensemble

Table 8: Performance Metrics of XGBoost-RF-SVM Ensemble Variants across Different Train/Test Split

Model Variant	Dataset	Train/Test Split	Accuracy (Acc)	Precision (Prec)	Recall (Rec)	F1-Score	Specificity (Spec)	AUC-ROC
XGB + RF + SVM	Raw	80/20	0.9737	1	0.9286	0.963	1	0.9947
XGB + RF + SVM	Balanced	80/20	0.979	1	0.9577	0.9784	1	0.998
XGB + RF + SVM	Increased Size	80/20	0.98	1	0.96	0.9796	1	0.9973
XGB + RF + SVM	Tuned	80/20	0.975	1	0.95	0.9744	1	0.9971
XGB + RF + SVM		90/10	0.99	1	0.98	0.9899	1	0.998
XGB + RF + SVM		80/20	0.98	1	0.96	0.9796	1	0.9973
XGB + RF + SVM		70/30	0.9734	1	0.9467	0.9726	1	0.9943
XGB + RF + SVM		60/40	0.9725	0.9896	0.955	0.972	0.99	0.9949

4.4 Performance Analysis

Impact of Data Balancing

Across all models, SMOTE balancing consistently enhanced recall by increasing the minority class (malignant). This was true. SVM polynomial's recall increased by nearly 30% from raw data to balanced data, with a 98.59% increase. The trade-off was a slight decrease in accuracy, as more false positives were produced with balanced data.

Impact of Dataset Size

In general, the dataset with more samples (500 samples per class) was found to be more stable and reliable than raw data. The variance of models trained on larger data was lessened when using different random seeds and split ratios.

Impact of Hyperparameter Tuning

By tuning the hyperparameters, there were slight enhancements in accuracy (usually 1-2 percentage points) when compared to the default parameters. SVM polynomial showed the greatest impact by tuning to improve both accuracy and recall.

Impact of Train-Test Split Ratio

The 90/10 split consistently produced the highest performance metrics across all models, as more training data led to better model generalization. The 60/40 split showed the lowest performance, with accuracy dropping by 1-2 percentage points compared to the 90/10 split. This trend was consistent across all model families, indicating that larger training sets benefit all algorithms equally.

4.5 Comparative Study and Discussion

The comparative study reveals several significant findings:

1. By utilizing ensemble methods (Stacking and Voting), it is clear that merging different learning algorithms is a wise decision, as they consistently perform better than individual classifiers.

2. Featuring an accuracy rate of 99% on the 90/10 split and perfect precision across most configurations, the XGBoost-RF-SVM Voting Ensemble is the overall winner.
3. Random Forest's consistency and minimal variation across various configurations make it a trustworthy baseline model.
4. SVM that incorporates polynomial kernel has the highest sensitivity to data conditions, with an accuracy of 88.6% on raw data and 98.60% on balanced data.
5. All models exhibit high specificity (with a precision of approximately 1.000), making them highly effective in correctly identifying benign cases. The ability to correctly identify malignant cases is the key factor that sets models apart, which involves recall.
6. Concavity, concave points, area, and perimeter are the most significant criteria for breast cancer classification found in SHAP analysis of both mean & worst measurements.

CONCLUSIONS

The project successfully implemented and evaluated four machine learning techniques for breast cancer detection using the Wisconsin Breast Cancer Dataset. The results showed that the XGBoost–Random Forest–SVM Voting Ensemble achieved the highest performance, reaching 99% accuracy and outperforming the conventional train-test split approach, which achieved 96.3% accuracy. The application of SMOTE for data balancing improved the recall of malignant cases, particularly for models such as Polynomial SVM that were affected by class imbalance. Among the different train-test ratios examined, the 90/10 split consistently produced the best results, indicating that increasing the amount of training data enhances model performance. Furthermore, SHAP analysis provided valuable interpretability by identifying concavity and area as the most influential features in predicting malignancy. Overall, all models demonstrated high specificity, making them effective in correctly identifying benign cases, while the primary difference in performance was their ability to accurately detect malignant tumors.

Future work can focus on exploring deep learning models such as CNNs to capture more complex patterns in breast cancer data, as well as applying advanced feature selection techniques to improve efficiency and accuracy. The study can also be extended to multi-class classification for identifying different breast cancer subtypes rather than only benign and malignant cases. Additionally, validating the models on larger and more diverse clinical datasets will help assess their real-world applicability and generalization. The developed Streamlit application can be further enhanced into a comprehensive clinical decision-support system by integrating explainable AI features and healthcare data platforms.

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APPENDICES

APPENDIX A: LIST OF ABBREVIATIONS

Abbreviation	Full Form
SVM	Support Vector Machine
RBF	Radial Basis Function
RF	Random Forest
SE	Stacking Ensemble
XGBoost	Extreme Gradient Boosting
SMOTE	Synthetic Minority Over-sampling Technique
SHAP	SHapley Additive exPlanations
WDBC	Wisconsin Diagnostic Breast Cancer
FNA	Fine Needle Aspirate
ROC	Receiver Operating Characteristic
AUC	Area Under the Curve
TP	True Positive
TN	True Negative
FP	False Positive
FN	False Negative
PCA	Principal Component Analysis
ANN	Artificial Neural Network

CNN Convolutional Neural Network

KNN K-Nearest Neighbors

APPENDIX B: EVALUATION METRICS FORMULAE

- $\text{Accuracy} = (\text{TP} + \text{TN}) / (\text{TP} + \text{TN} + \text{FP} + \text{FN})$
- $\text{Precision} = \text{TP} / (\text{TP} + \text{FP})$
- $\text{Recall (Sensitivity)} = \text{TP} / (\text{TP} + \text{FN})$
- $\text{F1-Score} = 2 \times (\text{Precision} \times \text{Recall}) / (\text{Precision} + \text{Recall})$
- $\text{Specificity} = \text{TN} / (\text{TN} + \text{FP})$
- AUC-ROC = Area under the curve of TPR vs FPR across all thresholds

APPENDIX C: SYSTEM SPECIFICATIONS

- Operating System: Windows 11
- Processor: Intel Core i-series
- RAM: 8GB+
- Python Version: 3.10+
- Development Tools: Visual Studio Code, Jupyter Notebook
- Key Libraries: scikit-learn, XGBoost, SHAP, imbalanced-learn, pandas, numpy, matplotlib, seaborn, streamlit

APPENDIX D: CD CONTENTS

The accompanying CD contains the following folders:

1. Presentation - PPT slides for the project
2. Documentation - Word file of the project report
3. Source Code - Full Python source code with notebooks
4. Program - Executable/runnable version of the project
5. Support - Third-party tools and runtime requirements
6. Help - User manual on how to run the project

REFLECTION OF THE TEAM MEMBERS ON THE PROJECT

Team Reflection

We were taught that systematic experimentation and comparative analysis are crucial components of machine learning projects as a team. During this project, we came to the realization that no single model is optimal, and every algorithm's value fluctuates depending on the data condition and evaluation metric. However, they are not uniform. The most effective way to detect breast cancer was through ensemble methods, which involve multiple classifiers. Additionally, we learned the importance of data preprocessing techniques like SMOTE in handling class imbalance and how hyperparameter tuning helps optimize model performance.

Individual Reflections

Rajaram Parida: As the team leader, I learned how to coordinate work across different models and ensure consistency in experimental conditions. Working with SVM and the Voting Ensemble helped me grasp the basics of margin-based classifiers and ensemble learning. Implementing SHAP explanations was especially rewarding because it made the models easier to understand and trust.

Saswat Parida: Through this project, I gained hands-on experience in data preprocessing and Random Forest classification. I learned how feature scaling and data augmentation affect model performance. Comparing Gini and Entropy criteria enhanced my understanding of decision tree-based algorithms.

Manpreet Mohanty: Working on Stacking Ensemble models taught me about meta-learning and how different base classifiers can work together. The hyperparameter tuning experiments revealed that even small changes in parameters can have a big impact on results. I also learned to recognize the balance between model complexity and performance.

Adyasha Biswal: My focus on result validation and comparative analysis helped me see the importance of using multiple evaluation metrics instead of just relying on accuracy. I learned that visualizations like confusion matrix heatmaps and ROC curves provide insights that numbers alone cannot express.

Subhashree Mohapatra

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